

Basic Concepts: Frequent Itemsets (Patterns)

- An itemset (or a pattern) X is *frequent* if the support of X is no less than a *minsup* threshold σ
- Let $\sigma = 50\%$ (σ : *minsup* threshold) for the given 5-transaction dataset



TID	Items bought
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
3	Beer, Diaper, Eggs
4	Nuts, Eggs, Milk
5	Nuts, Coffee, Diaper, Eggs, Milk

- All the frequent 1-itemsets:
 - Beer: 3/5 (60%); Nuts: 3/5 (60%); Diaper: 4/5 (80%); Eggs: 3/5 (60%)
- All the frequent 2-itemsets:
 - {Beer, Diaper}: 3/5 (60%)
- All the frequent 3-itemsets?
 - None

- Why do these itemsets (shown on the left) form the complete set of frequent k -itemsets (patterns) for any k ?
- **Observation:** We may need an efficient method to mine a complete set of frequent patterns